Overnight public information, order placement, and price discovery during the pre-opening period

Fariborz Moshirian[a], Huong Giang (Lily) Nguyen[b,*], Peter Kien Pham[a]

[a]*Australian School of Business, University of New South Wales, Sydney, NSW 2052, Australia*
[b]*La Trobe Business School, La Trobe University, Melbourne, VIC 3086, Australia*

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Abstract

This study examines the impact of corporate news announcements released overnight on price discovery during the pre-opening period in the Australian Securities Exchange. Our results suggest that the presence of these announcements increases the efficiency of indicative opening prices and that the intensity of these announcements significantly influences the aggressiveness of pre-opening orders. Using earnings announcements to compare the speed of price adjustments in response to overnight and daytime information of a homogeneous type, we find that prices respond immediately to overnight news upon the commencement of trading, whereas adjustments based on trading-hours news tend not to be instantaneous. Overall, our evidence highlights the important role of the pre-opening period in price discovery and the prospect of further enhancing this role by timing the release of public information to occur during non-trading hours.

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* Corresponding author. Tel.: +61 3 94793971; fax: +61 3 94791654.
E-mail addresses: f.moshirian@unsw.edu.au (Fariborz Moshirian), lily.nguyen@latrobe.edu.au (Huong Giang (Lily) Nguyen), peter.pham@unsw.edu.au (Peter Kien Pham).

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1. Introduction

It has been well established that public information in the form of continuous corporate disclosure is vital for the efficient functioning of capital markets[1]. In this paper, we study the impact of overnight corporate announcements on price discovery and order placement activities during the pre-opening period in the Australian Securities Exchange (ASX). Our study is motivated by the increasing tendency of corporate announcements to be made during the non-trading close-to-open period (Berkman and Truong, 2009)[2]. According to Doyle and Magilke (2009), this tendency reflects a desire to give the market more time to assimilate the information. Although prior studies often invoke information arrivals to explain how stock prices are established during the pre-opening period[3], few have examined the direct impact of overnight news announcements on order placement activities during this period and on the discovery of opening prices, especially in the context of an order-driven market.

An important feature of our analysis is the ability to obtain the exact timing of a news release when it first arrives at the market through an official and central distribution channel, i.e., the ASX. The corporate disclosure regulations in Australia (The ASX Listing Rule 3.1

[1] See Healy and Palepu (2001) and Verrecchia (2001) for a review.

[2] For example, Patell and Wolfson (1984) find that 19% of the earnings announcements in their sample of US firms in the late 1970s were released after the market had closed. Berkman and Truong (2009) find that the percentage of after-hours earnings announcements in their sample of US firms increased from 42% in 2000 to 49% in 2004.

[3] See, for example, Francis, Pagach, and Stephan (1992), Greene and Watts (1996), Biais, Hillion, and Spatt (1999), Cao, Ghysels, and Hatheway (2000), Masulis and Shivakumar (2002), Davies (2003), Brooks, Patel, and Su (2003), Comerton-Forde and Rydge (2006), Barclay and Hendershott (2008), Biais, Bisièsre, and Pouget (2009), and Kuk, Liu, and Pham (2011), among others.

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and Corporations Act 2001, Section 674)[4] ensure that the ASX is the first outlet of news announcements released to the public. By contrast, in existing studies, the date and time of news announcements are retrieved from secondary sources, including news wire services (e.g., Reuters and Bloomberg) and the financial press (through the Factiva database). Our results show that news announcements that arrive after the closing time of a trading day and before the opening time of the next trading day improve the efficiency of indicative opening prices[5]. Overnight information appears to be incorporated into prices upon trading more quickly than information released during normal trading hours. Furthermore, the intensity of overnight news announcements is a significant determinant of order placement activities during the pre-opening period.

The first part of our study focuses on the impact of overnight news announcements on the efficiency of indicative opening prices. This issue stems from the mixed evidence in prior studies regarding whether orders submitted prior to the market opening reflect noise or information about asset values. On the one hand, one could argue that it is unlikely for pre-opening orders to be construed as serious and informative because such orders are non-binding and may not lead to any trading[6]. With a market design such as that of the ASX, traders can observe, submit, revise or cancel their orders at zero cost before the market opens. In an analysis based on a sample of firms in the ASX, Kuk, Liu, and Pham (2011) report that

[4] Source: http://www.asxgroup.com.au/asx-listing-rules-guidance-notes-and-waivers.htm and http://www.comlaw.gov.au/Details/C2012C00275.

[5] Indicative opening prices at any time during the pre-opening period are the prices at which a security would be traded if the market were to open at that time.

[6] Normally, there is no trading or order matching during the pre-opening period. However, the NASDAQ allows pre-market and after-hours trading. The NASDAQ pre-market trading occurs before the market opens and between 7:00 am and 9:30 am. (Source: http://www.nasdaq.com/newscontent/20090216/ understanding-pre-market-and-after-hours-stock-trading.aspx).

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strategic orders – orders submitted with limited intention of trading (i.e., orders that are subsequently cancelled or revised) – occurred during the pre-opening period for 45.6% of their sample firms. For an experimental market, Biais, Bisièsre, and Pouget (2009) find that the non-binding feature of the pre-opening period provides an opportunity for traders to place manipulative orders, which can, in turn, lead to distrust in pre-opening communication among traders and limit the gain from trades.

On the other hand, there is some evidence that pre-opening orders contain useful information about fundamental asset values. For instance, Cao, Ghysels, and Hatheway (2000) document that NASDAQ dealers use crossed and locked inside quotes to signal to other market makers the direction in which prices should move during the pre-opening period. Davies (2003) observes that, in the Toronto Stock Exchange, the majority of orders submitted during the pre-opening period are placed with a serious intention of being executed.

Biais, Hillion, and Spatt (1999) reconcile these contrasting findings by showing that pre-opening orders can reflect both noise and price discovery. Based on data from the Paris Bourse, they observe a considerable degree of noise in indicative opening prices during the early hours of the pre-opening period. However, as the market approaches opening, the efficiency of indicative opening prices tends to increase. Such evidence supports the hypothesis that there is an evolution in the environment for learning about asset values, which reflects the role of the pre-opening period in facilitating a convergence of prices toward equilibrium.

Our study is closely related to Biais *et al.* (1999) but differs by focusing specifically on the direct impact of overnight company news announcements on the efficiency of indicative opening prices. We argue that if the role of the pre-opening period in price discovery can be theoretically attributed to overnight information arrivals, then the dissemination of public

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information should reduce the information asymmetry surrounding asset values. The noise component in indicative opening prices should be diminished by the presence of such information and the market should discover the equilibrium price more efficiently. Our results confirm that the efficiency of the indicative opening prices increases in the presence of overnight news announcements, thus highlighting the role of such news in facilitating the price discovery process. This finding remains robust after controlling for the New York Stock Exchange (NYSE) Composite Index return on the previous trading day, which is our proxy for information arriving from overseas markets.

The second part of our study focuses on the impact of overnight news announcements on order placement activities during the pre-opening period. Regarding the normal trading period, the literature suggests that order aggressiveness is influenced by factors such as market depth, volatility, order size, and spread size (Biais, Hillion, and Spatt, 1995; Parlour, 1998; Ranaldo, 2004; Beber and Caglio, 2005; Hall and Hautsch, 2007; Cao, Hansch, and Wang, 2008; Fong and Liu, 2010; among others). However, it remains unclear whether the same dynamics apply to the pre-opening period given that orders may only be executed when the market opens. To date, there has been little empirical evidence on the factors that determine order placement activities during the pre-opening period.

In our analysis, we argue that order submission (revision) decisions made by traders during the pre-opening period depend not only on the market conditions at the time of order submission (revision) but also on the existing information set at that time. In turn, such decisions can influence the probability of order execution when the market opens. Thus, we expect pre-opening information arrivals, in addition to market depth, volatility, and order size, to determine order placement activities during the pre-opening period. Because no actual order matching occurs during the pre-opening period, we measure order placement activities using order aggressiveness, which reflects the probability for a submitted order to be

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executed if the market were to open at that time. Our results indicate that, in addition to the factors already documented in the literature, the intensity of overnight news announcements is a significant determinant of order aggressiveness during the pre-opening period.

Having established the role of the pre-opening period in incorporating overnight information arrivals into prices, the final part of our study presents a comparative analysis of price discovery associated with news announcements made during trading hours (daytime) versus non-trading hours (overnight)[7]. Given that the market may respond differently to different types of information, we restrict our analysis to a relatively homogeneous type of news: earnings announcements. This methodology follows a large body of literature on differences in price discovery in response to daytime and overnight news announcements (Patell and Wolfson, 1984; Francis, Pagach, and Stephan, 1992; Greene and Watts, 1996; Masulis and Shivakumar, 2002; Brooks, Patel, and Su, 2003; Gerlach and Lee, 2011). For instance, Greene and Watts (1996) find that for overnight earnings announcements, the opening trade on the NYSE impounds most of the price response. By contrast, the response to daytime earnings announcements is spread evenly over the first few post-announcement trades. Brooks *et al.* (2003) also observe that the market reacts quickly to news released overnight about unanticipated corporate events but takes longer to react to similar news released in the daytime. However, thus far, studies on the differences between daytime and overnight price discovery have focused on the US markets (the NYSE and the NASDAQ), which differ from the ASX in terms of market structure. Our study complements this body of literature by focusing on a purely order-driven market.

Our results show that stock prices adjust rapidly to earnings announcements released during the overnight period. In particular, most of the price adjustments in response to such announcements occur during the pre-opening period and the first 15-minute interval after the

[7] We thank the reviewer for suggesting this to improve our study.

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market opens. By comparison, for an earnings announcement released during the normal trading period, investors need up to an hour, on average, after the release of the news to evaluate and absorb the information content of the announcement, and the price fluctuations after such an announcement tend to be larger. Thus, the pre-opening period appears to provide investors with an important time lag to evaluate overnight information and such information is incorporated into stock prices more quickly upon the commencement of trading.

Overall, we contribute to the literature by providing new evidence for the role of overnight news announcements in facilitating the price discovery process and influencing order placement activities during the pre-opening period in a purely order-driven market such as the ASX.

The remainder of this paper is organized as follows. In Section 2, we present institutional details. In Section 3, we provide a description of the data. In Section 4, we report and discuss the main results. Finally, in Section 5, we provide our concluding remarks.

2. Institutional details - The ASX opening procedures

The ASX operates under the fully computerized Stock Exchange Automated Trading System (SEATS) with the trading rules of an electronic limit order market. For a trading day, the pre-opening period starts from 7:00 am to 10:00 am. The opening period lasts from 10:00 am to 10:09 am. During the opening period, securities are opened for trading in five groups according to the starting letter of their ASX codes. Between 10:09 am and 4:00 pm, the normal trading period takes place continuously for all stocks. The pre-closing period occurs from 4:00 pm to 4:10 pm and the closing period occurs from 4:10 pm to 4:12 pm.

During the pre-opening period, there is no trading; however, traders can submit and revise orders for possible execution at the opening time. Market depths and price levels are

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updated with these orders. Because orders are not executed until the market opens, a situation may occur in which the highest bid is equal to (locked quotes) or greater than (crossed quotes) the lowest ask. If there are either locked or crossed quotes (i.e., the two sides of the order book overlap), the opening price is determined using the ASX's opening price algorithm, which follows four principles to calculate a transaction price that allows for the maximum quantity of shares to exchange ownership. Principle 1 involves selecting a price that maximizes the executable trading volume. If there is a range of prices at which this occurs, the algorithm considers Principle 2, which selects, within this range, the price that minimizes the surplus of cumulative buy quantity over cumulative sell quantity. If there is still more than one price by which this surplus is minimized, the algorithm applies Principle 3, which selects the highest (lowest) price if the surplus is on the buy (sell) side. If both buy and sell surplus volumes are present at these prices, the algorithm applies Principle 4, which simply sets the previous closing price as the opening price (see Appendix A for further details).

3. Data and variables

The data for this study come from the Securities Industry Research Centre of Asia-Pacific (SIRCA) and cover constituents of the S&P/ASX 100 index for the period lasting from July 1, 2005 to June 30, 2006. We use only one year of data because the sample size of order book data is prohibitively large. The S&P/ASX 100 index comprises the 100 largest stocks in the ASX based on market capitalization. As Madhavan and Panchapagesan (2000) suggest, it is not practical to trade all securities in the pre-opening session, especially those that are thinly traded in a fully automated call market because relatively small imbalances can generate prices that are not economically meaningful. For this reason, we select the constituents of the S&P/ASX 100 index because these securities are actively traded during

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normal trading hours. Details regarding trades, orders, market depths, and company news announcements are obtained for each of the sample stocks.

3.1. Overnight news announcements

The public news database of the ASX differs from those of many other stock exchanges in that its listing rules ensure that all company news announcements are released through it first before any other financial news outlets. In particular, listed firms are required to submit their announcements to the ASX before releasing the information through their own media (or other public outlets). Upon submission, each announcement is released immediately and centrally by the ASX to the public with a time stamp.

In this study, any company news announcement released to the ASX after the market closes and before the market opens on the next trading day is considered an overnight news announcement. When examining the impact of overnight news announcements on the efficiency of indicative opening prices, we use a news dummy variable (D_news_t) for each one-minute interval between 7:00 am and the market opening time. D_news_t is equal to one if at least one announcement arrives between the closing time of the previous trading day and that one-minute interval and zero if otherwise. In this analysis, the dummy variable D_news_t is used because we only need to compare the role of the pre-opening period in price discovery across two regimes, one with overnight news announcements and the other without overnight news announcements.

However, when examining the impact of overnight news announcements on order placement activities during the pre-opening period, we need another measure based on the intensity of overnight news announcements up to a specific order event (i.e., at time t). This variable (N_News_t) is calculated as the cumulative number of all announcements released between the closing time of the previous trading day and the time of order submission (revision) (time t). The rationale for using the cumulative number of announcements reflects

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the fact that, until the commencement of trading, any order submitted at a given time is likely to be driven by consideration for all market available information accumulated until that time.

In our study, there are a total of 10,417 news announcements from the sample firms, of which 5,222 announcements (50.13%) are released during the overnight period and 5,195 announcements (49.87%) are released during the normal trading period. For each firm, the daily average number of overnight news announcements is 0.23 whereas the daily average number of daytime news announcements is 0.22. We report the frequency distribution of news announcements across half-hour intervals in Figure 1, which shows that the news announcements are released primarily between 9:00 am and 10:00 am and between 3:30 pm and 5:00 pm.

Insert Figure 1 here

3.2. Indicative opening prices and stock returns

To study price discovery and order activities during the pre-opening period, we replicate the ASX opening algorithm (as discussed in Section 2) to determine indicative opening prices. The indicative opening price (P_t) at time t during the pre-opening period is the price at which traders would transact if the market were to open at that time. We evaluate how overnight information is incorporated into stock prices based on the evolution of indicative opening prices. In other words, we examine the correlation between indicative returns and close-to-close returns. In particular, we define the indicative return (R_Ind_t) at a given time (in one-minute intervals between 7:00 am and the opening time) as the logarithmic return of the indicative opening price at that time over the closing price of the previous day. The close-to-close return (R_Close) is the logarithmic return of the closing price of a trading day over the closing price of the previous trading day.

Insert Table 1 here

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Table 1 provides summary descriptive statistics on indicative and close-to-close returns for the entire sample, the subsample of days without any overnight news announcements (no-news sample), and the subsample of days with at least one overnight news announcement (news sample). The mean indicative return is 0.018% for the no-news sample and 0.006% for the news sample. The mean close-to-close return is 0.046% for the no-news sample and 0.062% for the news sample.

3.3. Order placement or order aggressiveness

To study the impact of overnight news announcements on pre-opening order placement strategies, we focus on order aggressiveness. Because order execution does not occur until the market opens, we argue that traders' decisions on where to place their orders are linked to the probability that their orders will be executed at the opening time. Although there is a time gap between the time of order submission (revision) and the opening time, pre-opening traders are inclined to infer that, on average, the more aggressive an order is at the time of submission (revision), the higher the probability for such an order to be executed at the opening time. Thus, we define order aggressiveness at time *t* during the pre-opening period based on the relative location of the order in the order book compared to the best bid, best ask, and the indicative opening price using the following procedures.

We first classify orders into five types. *Submit* orders are those that are entered into the system for the first time. The rest are *Revise* (revisions of previously submitted) orders, of which there are four categories. *Forward* refers to orders that are revised to a more aggressive price level (higher prices for buy orders or lower prices for sell orders). *Same* refers to orders that are revised only in terms of order size (either increase or decrease) but not order prices. *Backward* refers to orders that are revised to a less aggressive price level (lower prices for buy orders or higher prices for sell orders). *Delete* are orders that are deleted before the market opens.

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We then classify orders into three levels of aggressiveness (sub-categories) by comparing the price of each order with the best bid, best ask, and the indicative opening price at the time of order submission (revision). The least aggressive orders (Level 1) are buy (sell) orders with prices that are lower (higher) than the best ask (bid) prices. Level 2 orders are buy (sell) orders with prices that are higher (lower) than or equal to the best ask (bid) prices and lower (higher) than the indicative opening price. The most aggressive orders (Level 3) are buy (sell) orders with prices that are higher (lower) than or equal to the indicative opening price. Figure 2 provides a more detailed illustration of this classification.

Insert Figure 2 and Table 2 here

In Table 2, we present summary statistics on order aggressiveness. Two features are evident. First, Level 3 orders are dominant, accounting for over 52% of all order types (excluding the deleted orders), and the Level 1 sub-category is the least prevalent of all order types. Second, for both the Level 2 and 3 orders, the proportion of orders submitted during the last half-hour interval (from 9:30 am to the opening time) dominates that of the orders submitted during earlier times of the pre-opening period. This pattern suggests that, on average, pre-opening orders become more aggressive as the market moves closer to opening.

3.4. Control variables

Based on prior work in the literature on order aggressiveness (e.g., Biais *et al.*, 1995; Ranaldo, 2004), we use the following control variables in our analysis of order placement activities during the pre-opening period.

Market Depth: We calculate the depth of the bid (ask) side as the natural logarithm of the total number of shares outstanding in the bid (ask) side of the order book. As Parlour (1998) shows, the probability of order execution depends on the size of the order book and the beliefs that traders hold about further order arrivals. An increase in market depths on the

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buy (sell) side reduces the probability of executing buy (sell) limit orders and, thus, encourages traders to submit more aggressive buy (sell) orders. This argument should also apply to the pre-opening period. Specifically, an increase in market depths on the buy (sell) side can reduce the probability of executing buy (sell) orders at the opening time because traders on the same side face greater competition. To increase the probability of order execution, traders may have to submit or revise their orders at more aggressive prices. As a result, the market depth of both the same side ($Depth_Same$) and the opposite side $(Depth_Oppo)$ is expected to be significant determinants of order aggressiveness during the pre-opening period.

$Vola:$ Volatility is measured as the absolute value of the close-to-close return from the previous trading day. There is a well-established literature documenting that stock return volatility is directly related to information arrivals (e.g., Clark, 1973). As such, volatility from the previous trading day may serve as a proxy for the intensity of aggregate information arrivals and the level of uncertainty about stock values that persist from the previous trading day. Therefore, we expect the volatility of the previous trading day to be a determinant of order aggressiveness during the pre-opening period.

$Size:$ We construct this variable as the natural logarithm of the number of shares quoted in each order. Prior literature suggests that order size may convey information about asset values. For instance, Easley and O'Hara (1987) show that informed traders are likely to use large trades instead of small trades. By contrast, Barclay and Warner (1993) show that informed traders may camouflage their private information and split their large trades into medium trades. Because actual trading does not occur during the pre-opening period and any orders submitted are non-binding, traders may be more likely to use a large order as a means of impacting opening prices. Thus, we expect order size to be another determinant of order aggressiveness during the pre-opening period.

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Cross: We define this variable as the absolute value of the difference between the best bid and the best ask at the time of order submission or revision. As shown in Cao *et al*. (2000), NASDAQ dealers use crossed inside quotes to signal to the other market makers the direction of price movement. Thus, we expect cross size to also be a significant determinant of order aggressiveness during the pre-opening period.

4. Results and discussions

4.1. Overview of market activities during the pre-opening period in the ASX

We provide an overview of market activities during the pre-opening period in Tables 3, 4, and 5. Table 3 presents the proportion and size of orders. Table 4 presents the characteristics of an overlapping market, in which the best bid quote is equal to or higher than the best ask quote. Table 5 presents the weighted price contribution, which is the ratio of the price change during the pre-opening period to the total daily price change.

Insert Table 3 here

In Panel A of Table 3, we report the proportion and size of orders across half-hour intervals during the entire pre-opening period and across ten-minute intervals during the last hour before the market opens. Order submission/revision activities are most active in the last thirty minutes (and especially in the last ten minutes) before the market opens. Specifically, the proportion of orders from 9:30 am to the opening time accounts for 63.61% of all orders during the pre-opening period, and the proportion of orders from 9:50 am to the opening time accounts for 73.09% of all orders received between 9:00 am and the opening time. This finding suggests that traders generally do not submit orders early but wait until the market approaches opening time. This pattern aligns with the evidence documented in Davies (2003) for the Toronto Stock Exchange, in Biais *et al*. (1999) for the Paris Bourse, and in Comerton-Forde and Rydge (2006) for the ASX.

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In Panel B of Table 3, we document that the proportion of newly submitted orders (49.81%) is similar to that of revised orders (50.19%) over the entire pre-opening period. However, from 9:30 am to the opening time, newly submitted orders account for only 25.26% while revised orders account for 38.35% of all orders over the entire pre-opening period (making up 63.61% as reported in Panel A). In addition, if we compare the order size between 7:00 am and the opening time with that between 9:30 am and the opening time, the size of new submitted orders decreases and the size of revised orders increases as the market approaches opening time.

Two potential explanations exist for such changes in the proportion and size of newly submitted and revised orders. First, this pattern is consistent with a hypothesis related to the investor learning environment, which suggests that investors submit and revise their orders in response to changes in market conditions due to an increasing flow of new information arrivals during the pre-opening period. Second, there is a possibility of order manipulation, that is, when the market is far away from opening, many new orders are placed with the intention of attracting market activity and influencing prices. However, when the market is just about to open, these orders are quickly revised or withdrawn.

Insert Tables 4 and 5 here

In the absence of trading, an overlapping market is another defining characteristic of the pre-opening period. Table 4 shows that negative spreads (or crossed quotes) are dominant during the observed pre-opening periods of our sample firm-days (95.3%). Positive and zero spreads account for only 3.06% and 1.65%, respectively, of all pre-opening quotes. This pattern of overlapping buy and sell orders for the ASX is similar to that observed in Davies (2003) regarding the pre-opening session of the Toronto Stock Exchange. This result highlights the differences between the opening session in an order-driven market, which is

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dominated by locked and crossed quotes, and those in a quote-driven market, in which locked and crossed quotes account for only approximately 36% (Cao *et al*., 2000).

In Table 5, we report the weighted price contribution (WPC) of the pre-opening period, which is the ratio of the price change during this period to the total daily price change. As shown in Cao *et al.* (2000) and Barclay and Hendershott (2003, 2008), WPC provides a measure of the magnitude of price discovery in a given period. WPC statistics are reported for three periods: between the closing time on the previous day and 9:30 am, between 9:30 am and the opening time, and between the opening time and the closing time.

As Table 5 shows, the WPC for the period between the closing time on the previous day and the opening period accounts for 31% of the total daily price change, whereas the WPC for the period between opening and closing times accounts for 69% of the total daily price change. When the period between the closing time on the previous day and the opening time is divided into two sub-periods – from the closing time on the previous day to 9:30 am and from 9:30 am to the opening time, the former sub-period contributes 17% and the latter sub-period contributes 14% to the total daily price change. These results are similar to findings documented in Barclay and Hendershott (2003, 2008) and Cao *et al*. (2000) for the NASDAQ pre-opening session and suggest that the pre-opening period, especially the last half hour before the market opens, is an important factor influencing price discovery in the ASX.

4.2. Overnight news announcements and the efficiency of indicative opening prices

To study the price efficiency of the pre-opening period, we follow Biais *et al.* (1999), who develop an econometric framework to analyze whether the evolution of indicative opening prices reflects investor learning or simply noise. According to the authors, if the pre-opening period efficiently incorporates information about the equilibrium value of the asset

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(V), then the indicative opening price (P_t) at pre-opening time t should reflect the market's conditional expected value of the asset $E(V|I_t)$ given the information set at that time (I_t). The implication is that if the equilibrium asset value at the start of the pre-opening period is $E(V|I_0)$, then the difference between the indicative opening price and this starting value [P_t - $E(V|I_0)$] should predict the change in the underlying value of the asset [V-$E(V|I_0)$]. To operationalize this argument in a regression analysis, they assume that the closing stock price at the end of a trading day reflects the true value of the assets of the firm on that day[8], yielding the following regression:

$$R_Close = \alpha + \beta * R_Ind_t + \varepsilon_t, \quad (1)$$

where R_Close is the close-to-close return, which is equal to the natural logarithm of the ratio of the closing price on a given trading day to that on the previous trading day and represents the change in the equilibrium value of the asset [V-$E(V|I_0)$]. R_Ind_t is the indicative return, which is equal to the natural logarithm of the ratio of the indicative opening price at time t to the previous close price and represents the difference between the indicative opening price at time t and the starting equilibrium value [P_t-$E(V|I_0)$].

To examine the impact of overnight news announcements on the efficiency of indicative opening prices, we extend the methodology used in Biais *et al.* (1999) by including an interaction term between the news dummy and the indicative return (D_News_t*R_Ind_t). This variable captures the incremental impact of overnight news announcements on the efficiency of indicative opening prices between one-minute intervals with overnight news announcements and those without overnight news announcements. In particular, we estimate the following regression:

[8] Several other studies, such as Ciccotello and Hatheway (2000) and Comerton-Forde and Rydge (2006), assume that the true value of the asset is the midpoint of the best quote prevailing in the market at another time such as 11:00 am. Their results are similar to Biais *et al*. (1999).

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$$R_Close = \alpha + \beta * R_Ind_t + \gamma * D_News_t * R_Ind_t + \varepsilon_t, \quad (2)$$

where the news dummy variable (D_News_t) equals one if at least one announcement arrives between the closing time of the previous trading day and time *t* and zero if otherwise. All remaining variables are the same as in the regression (1).

An econometric issue in this regression analysis is that the amount of noise in indicative opening prices can also decline with the approach of the market opening time. To account for such non-stationarity of the distribution of indicative opening prices, we follow Biais *et al*. (1999) and estimate the regression (2) using an unbiased regression procedure, in which we run one regression across days for each one-minute time interval during the pre-opening period. This exercise is repeated stock by stock, with the average coefficient of β for each minute reported across all sample stocks. If the coefficient estimate on R_Ind_t (β) is positive on average and statistically significant, then there is support for the argument that indicative opening prices are efficient. If the coefficient estimate on D_News_t*R_Ind_t (γ) is positive on average and statistically significant, then there is support for the hypothesis that overnight news announcements increase the efficiency of indicative opening prices. For comparison purposes, we also apply the same procedure to transaction prices for all post-opening one-minute intervals between 10:00 am and 10:30 am.

We present the estimation results in Figure 3, which shows graphically the mean coefficient estimates of the regression (2). As there is no discernible pattern in these estimates for the intervals preceding 9:30 am, we present only the mean coefficient estimates for the period from 9:30 am to 10:30 am. The solid line is the mean coefficient estimate of β and the dashed line is the mean coefficient estimate of γ. All estimates of β are statistically significant at the 5% level for the intervals after 9:30 am. All estimates of γ are statistically significant at the 5% level for the intervals from 9:30 am to the opening time.

Insert Figure 3 here

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Results in Figure 3 show that indicative opening prices become more efficient as the market approaches the opening time. After 9:30 am, the mean coefficient estimate of the indicative return variable (β) stays below 1.0 but there is a general upward trend in its value. Between 9:30 am and 9:50 am, β reaches approximately 0.4 to 0.5. From 9:50 to the opening time, there is a sharp increase in the value of β, which stays at approximately 0.9 at the opening time. The value continues to increase and reaches 1.0 around fifteen minutes after the market opens. In additional robustness checks that are not graphically depicted in Figure 3, we find that the estimate of β stays above 1.0 at most times after 10:30 am during the trading period and then gradually declines to 1.0 as the market approaches the closing time.

These results suggest that the efficiency of indicative opening prices begins to improve after 9:30 am and increases sharply during the last ten minutes before the market opens. Consistent with Biais *et al*. (1999), this evidence suggests that a large number of traders wait until the market approaches the opening time before taking active part in order activities. Thus, we find support for the hypothesis that, during the pre-opening period in the ASX, learning about the equilibrium valuation of security prices occurs as the market approaches the opening time, resulting in better price discovery and a convergence of prices toward equilibrium.

More importantly, the results in Figure 3 also show that the mean coefficient estimate of the interaction term between the news dummy and the indicative return (γ) is positive, as illustrated by the dashed line, which stays well above zero. This finding indicates that, with the arrival of overnight news announcements, the learning environment during the pre-opening period becomes more active and contributes to an increase in the predictability of the final closing price (as a proxy for the market equilibrium price) based on indicative opening prices. Thus, variations in the efficiency of the price discovery process appear to be driven by the arrival of public information, further suggesting that the pre-opening period does not

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simply contain noise but also plays an important role in incorporating information into prices through investor learning.

We perform several robustness checks on the above finding. In particular, we consider a source of public information other than overnight company news announcements, namely, the performance of other overseas markets. The trend of integration and globalization of financial markets around the world, the significant interdependence of world market movements, and the information leadership role of the US market in particular, have all been well documented (Arshanapalli and Doukas, 1993; Brooks and Negro, 2006; Wang and Yang, 2011; among others).

The geographical location of Australia means that the pre-opening period of the ASX is influenced most directly by trading in the US markets. In particular, the trading hours of the markets in Europe and the US occur after the closing time of the ASX. By contrast, major Asian markets, such as the Tokyo Stock Exchange and the Hong Kong Stock Exchange, open at the same time or after the opening of the ASX. These trading time differences suggest that the ASX pre-opening period receives flows of information about activities in the European and US markets that coincide with the ASX non-trading hours. However, because the European markets close earlier than the US markets, US markets are more likely to pose a direct impact on the pre-opening period of the ASX. Indeed, based on daily index data (obtained from the Thomson Reuters Tick History database) for the NASDAQ Composite, the NYSE Composite, the S&P 500 (US), the FTSE 100 (London), the CAC 40 (Paris), the DAX (Frankfurt), the Nikkei (Tokyo), and the Hang Seng (Hong Kong), we find our sample stocks to be more highly correlated with the US indices than other international indices. Because the highest correlation tends to occur with the NYSE Composite index, we use the

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NYSE Composite index return as a proxy for the flow of information arriving from the overseas markets[9].

We then modify the regression (2) to account for the NYSE Composite index return as follows:

$$R_Close = \alpha + \delta * R_NYSE + \beta * R_Ind_t + \gamma * D_News_t * R_Ind_t + \varepsilon_{t,} \quad (3)$$

where R_NYSE is the return of the NYSE Composite index from the previous trading day.

Figure 4 presents the estimation results for the regression (3). As with Figure 3, we present the mean coefficient estimates for the period from 9:30 am to 10:30 am only. The solid line is the mean coefficient estimate of β, the dashed line is the mean coefficient estimate of γ, and the dotted line is the mean coefficient estimate of δ. All estimates of β are significant at the 5% level for the intervals after 9:30 am. All estimates of γ and δ are significant at the 5% level for the intervals from 9:30 am to the opening time.

Insert Figure 4 here

There are several notable findings depicted in Figure 4. First, the coefficient estimate of R_NYSE (δ) is strictly positive, as shown by the dotted line staying well above zero. This result suggests that information arriving from the overseas market contributes to the price discovery process. Second, the coefficient estimate of R_Ind_t (β) starts to increase after 9:45am, as shown by the upward trend in the solid line. This result shows that after the

[9] After-hours trading periods (4:00 - 8:00 pm) in the US markets overlap with the ASX pre-opening period. However, after-hour trades tend to occur for only certain types of stocks with limited trading volume. For instance, Barclay and Hendershott (2003) report that only 4% of the NASDAQ trading volume occurs after hours. The NASDAQ shows after-hour trades on its public page only if the stock is a component of the NASDAQ-100 Index. Thus, we assume that the ASX pre-opening traders learn more from the overall performance of the overseas markets during the trading-hours period than from the after-hours period.

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overseas market information is incorporated, we still observe results similar to those presented in Figure 3. Third, we find that the coefficient estimate of D_News_t*R_Ind_t (γ) remains positive, indicating that even after incorporating information from overseas markets, the existence of local overnight news announcements still improves price efficiency during the pre-opening period.[10]

As another robustness check, we control for the potential impact of trading-hours (daytime) news announcements on the pre-opening price efficiency of a given trading day. We consider both trading-hours news announcements released on the same trading day and those released on the previous trading day in two unreported additional tests. The first test concerns whether trading-hours news announcements on the same trading day dilute the effect of overnight news announcements on the efficiency of indicative opening prices on that trading day. This involves splitting the sample into firm-days with and without trading-hours news announcements. We then examine the efficiency of the indicative opening prices during the pre-opening period for the two sub-samples by re-estimating the regression (2). The results show no significant difference in the coefficient estimates between these two sub-samples. Thus, the presence of trading-hours announcements on the same trading day does not appear to dilute the effect of overnight announcements on the efficiency of indicative opening prices.

The second test involving trading-hours news announcements examines whether the presence of these announcements in the previous trading day has an information spillover effect on the pre-opening period of a given day. A trading-hours dummy ($Pre_Trading_News$), equal to one if trading-hours news announcements are released during the previous trading day and zero if otherwise, is used as a proxy for trading-hours

[10] In an unreported robustness check, we incorporate in the regression (3) an interaction term between *R_NYSE* and R_ind_t. Again, we do not find any significant change in our results with respect to the other variables.

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news announcements on the previous day. Then, we interact the trading-hours dummy with the indicative return variable and modify the regression (2) to incorporate this interaction ($Pre_Trading_News * R_Ind_t$). This interaction term captures the incremental correlation between indicative returns and close-to-close returns (or the incremental efficiency of the indicative opening prices) if news announcements were made during trading hours on the previous day. However, the coefficient estimates of this interaction are, on average, not significantly different from zero, whereas the other estimates are qualitatively similar to those reported for the original specification of the regression (2). These results suggest that trading-hours news announcements from the previous trading day do not have an information spillover effect on the pre-opening period of a given trading day. The most plausible explanation, which is consistent with Chordia, Roll, and Subrahmanyam (2005), is that price adjustments in response to trading-hours news announcements tend to occur rapidly within the previous trading day. The second robustness check indicates that our result is robust against the potential impact of trading-hours news announcements. For brevity, we do not report the detailed results of both of the above robustness checks .

4.3. Overnight news announcements and order placement activities

Having established that overnight news announcements contribute to price efficiency by improving the learning environment of pre-opening investors, in this section we examine the effect of overnight news announcements on order activities. Following the literature on the determinants of order aggressiveness (Biais *et al.*, 1995; Parlour, 1998; Griffiths *et al.*, 2000; Ranaldo, 2004; Beber and Caglio, 2005; Hall and Hautsch, 2007; Cao *et al.*, 2008; Fong and Liu, 2010; among others), we use an ordered probit model as follows:

$$Z_t = \sum \beta_i X_{i,t-1} + \varepsilon_t \tag{4a}$$

$$Y_t = 1 \; if \; Z_t = (-\infty, \mu_1]; \; Y_t = 2 \; if \; Z_t = (\mu_1, \mu_2]; \; Y_t = 3 \; if \; Z_t = (\mu_2, +\infty), \tag{4b}$$

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where Z_t is the latent continuous variable related to the discrete variable (Y_t), which reflects the levels of order aggressiveness. In particular, the Y_t = 1 category contains the least aggressive orders (Level 1), the Y_t = 2 category contains the orders that are intermediate in aggressiveness (Level 2), and the Y_t = 3 category contains the most aggressive orders (Level 3). The parameters μ_1 and μ_2 are the partition points to be estimated and the error term ε_t is assumed to follow the standard normal distribution. X_i is the set of explanatory variables including the intensity of overnight news announcements (N_News), the return of the NYSE index from the previous trading day (R_NYSE), market depth ($Depth_Same$) and $Depth_Oppo$), stock return volatility ($Vola$), order size ($Size$), and cross size ($Cross$). Our main variable of interest is N_News, which we expect to positively influence order aggressiveness in the pre-opening period.

In Table 6, we report the regression results of estimating (4a) and (4b). Panel A reports the averages of the coefficient estimates and *t*-statistics. Panel B reports the marginal effects of the explanatory variables, which are calculated by using the sample average of individual marginal effects and scaled by 100.

Insert Table 6 here

As reported in Panel A, the coefficient estimates of N_News are all statistically significant for all order types. For newly submitted orders (*Submit*), the marginal effects of N_News (Panel B) are positive for the most aggressive orders (Level 3) (0.38%) and negative for the least aggressive orders (Level 1) (-0.12%). These results suggest that the greater the number of overnight news announcements, the more aggressive the *Submit* orders. We observe the same pattern of marginal effects for two types of revision orders: those revised to more aggressive price levels (*Forward*) and those remaining at the same price (*Same*). These findings suggest that for the majority of orders (i.e., *Submit* orders and *Forward* and *Same* revision orders), the more overnight news announcements made, the more aggressive the pre-

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opening orders. In other words, traders do take into account news announcements arriving during the overnight period in determining their order aggressiveness. Thus, there is evidence to suggest that the opening period provides a learning environment for investors who aggressively submit orders to capitalize on new information.

It is important to note that we cannot rule out the possibility that overnight information arrivals facilitate market manipulation by some traders. In fact, in the case of *Backward* revision orders, the marginal effect of N_News is negative for the most aggressive orders (Level 3 *Backward* orders), indicating a negative effect of N_News on order aggressiveness. This finding suggests that in the case of a backward revision, overnight news announcements reduce the likelihood for an order to be executed. In the regression on *Delete* orders, the marginal effect of N_News is positive for the most aggressive orders (Level 3 *Delete* orders), indicating a positive effect of N_News on order aggressiveness. This finding implies that in the case of an order deletion, overnight news announcements increase the likelihood of deleting an aggressive order. Whereas these two results may reflect market learning, they are also consistent with another explanation involving market manipulation, as discussed in Kuk *et al.* (2011). Specifically, the overnight arrival of new information may create short-term uncertainty about stock prices and make certain traders more likely to employ the strategy of submitting an aggressive order and later revising it down (or cancelling it) with the purpose of manipulating the market. However, given that the relatively small proportion and average order size of *Backward* and *Delete* orders during the half hour before the market opens (see Panel B of Table 3), market manipulation does not appear to be a dominant aspect of the pre-opening period.

Turning to the other explanatory variables, we find the coefficient estimates of R_NYSE to be statistically significant at the 10% level or better, as shown in Column 2 of Panel A. This result suggests that information from the overseas markets influences order

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placement activities during the pre-opening period. In addition, the effect of market depth on order placement during the pre-opening period is similar to the one during normal trading hours. All coefficient estimates of $Depth_Same$ ($Depth_Oppo$) are statistically significant at the conventional levels, which are shown in Column 3 and Column 4 of Panel A. Furthermore, we can see that the marginal effect of $Depth_Same$ ($Depth_Oppo$) on Level 3 submit orders is 2.01% (-1.27%) in Panel B. These findings suggest that there is a positive (negative) relationship between market depth on the same (opposite) side and aggressiveness of the newly submitted orders, which is consistent with Parlour (1998).

The remaining variables – volatility, order size, and cross size – all influence order activities during the pre-opening period. For instance, the marginal effects of cross size ($Cross$) on Level 3 orders are negative for all types of orders. In other words, the greater the degree of overlap in the existing order book, the greater the likelihood that an equilibrium price has been reached and the less aggressive the incoming orders are. The marginal effects of order size ($Size$) on Level 3 orders are positive for *Submit* or *Forward* orders, which means that the larger the order size, the more aggressive the submitted and amended forward orders are.

4.4. Price discovery following daytime and overnight earnings announcements

Thus far, we have shown that the pre-opening period plays an important role in facilitating investor learning and incorporating overnight news into opening prices. However, our original methodology (the regression (2)) does not allow us to directly compare the efficiency of the price discovery process between the pre-opening period and the normal trading period. To do this, we need to account for fundamental differences between the two periods in terms of both order execution and the intensity (and type) of information arrivals. In particular, we adopt the event methodology used in the related studies (e.g., Greene and

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Watts, 1996; Masulis and Shivakumar, 2002; Brooks *et al.*, 2003; Gerlach and Lee, 2011) and examine how the market responds to a relatively homogenous type of information: earnings announcements. Our focus is on comparing the speed at which trading incorporates new information into prices when earnings announcements are released during the overnight period versus when earnings announcements are released during the normal trading period.

For all the stocks in our sample, we obtain the timing of the earnings announcements from SIRCA and analyst earnings forecasts and actual earnings per share from the Institutional Brokers Estimate System (I/B/E/S) summary tape for the period between 2003 and 2007. We also require the presence of trades and quotes three days before and three days after the event date. Table 7 reports the distribution of the earnings announcements in our sample. There are a total of 423 annual earnings announcements, with 319 announcements (75%) released during the overnight (including pre-opening) period (with only 5 announcements preceded by a trading halt) and 114 earnings announcements (25%) released during the normal trading period (with no announcements preceded by a trading halt).

Earnings announcements are interpreted as positive or negative based on analyst forecast errors (AFE). Following Greene and Watts (1996) and Livnat and Mendenhall (2006), we use analyst forecast errors as a proxy for the unexpected component in earnings figures released to the market. The AFE is calculated as the difference between the actual earnings estimate and the mean earnings forecast at the time of an earnings release, scaled by the absolute value of the mean earnings forecast. An earnings announcement with a positive (or negative) AFE is considered positive (or negative) earnings news. In terms of descriptive statistics, the mean (median) AFE for overnight earnings announcements is 0.003 (0.002) with a standard deviation of 0.14. The mean (median) AFE for daytime earnings announcements is 0.05 (0.01) with a standard deviation of 0.75. The summary statistics in

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Table 7 show that 60% (254 announcements) of the earnings announcements are positive and the remaining 40% are negative (179 announcements).

Insert Table 7 here

To assess the impact of daytime and overnight earnings announcements on the speed of price adjustments, we track the intraday returns on a portfolio with long positions in stocks with positive earnings news and short positions in stocks with negative earnings news on the event date. Following Greene and Watts (1996) and Masulis and Shivakumar (2002), we use a 15-minute event interval. Given that the ASX operates between approximately 10:00 am and 4:00 pm, we divide a trading day into 25 intervals that span both the pre-opening and trading periods. For each day, the first interval starts at the closing time of the previous trading day and ends at the opening time of a given trading day. This decision is necessitated by the requirement to apply a homogeneous return measurement setting to both daytime and overnight announcement events. The second interval starts at the opening time and ends at 10:15 am. All subsequent intervals are 15 minutes in duration. The last interval starts at 3:45pm and ends at the closing time of the given trading day.

For daytime earnings announcements, the event interval 0 is defined as the 15-minute interval containing the earnings announcement. For overnight earnings announcements, the event interval 0 starts at the closing time of the previous trading day and ends at the opening time of the trading day (or at the first trade if trading halts occur). All other event intervals are defined relative to the event interval 0. For each event interval, we calculate the excess return for each stock by taking the difference between its return and the S&P/ASX 100 index return, where the stock and index returns are based on the midpoints of their best bid and ask quotes at the end of each event interval. We then compute the returns of two equally weighted portfolios: one for overnight earnings announcements and the other for daytime earnings announcements.

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As reported in Table 8, for daytime earnings announcements, the portfolio returns are positive and statistically significant for the intervals 0, +1, +2, and +3, with the return for the interval 0 being highest at 0.10%. Returns for the other event intervals are almost all statistically insignificant. Thus, it appears that the market takes approximately four 15-minute intervals to absorb the information content of an earnings announcement. By contrast, portfolio returns in response to an overnight earnings announcement are positive and statistically significant for only the intervals 0 and +1. The majority of the price adjustments occur during the overnight period (interval 0), with the portfolio returns for the intervals after interval +1 (ranging from -0.03% to 0.03%) being statistically insignificant. In terms of magnitude, the portfolio return for interval 0 is also greater, on average, for overnight announcements (at 0.15%) than for daytime announcements (at 0.10%). Nonetheless, we cannot rule out the selection effect or the tendency for companies to release more material earnings announcements during the overnight period.

Insert Table 8 here

Figures 5 and 6 provide a visual representation of the cumulative portfolio returns for daytime and overnight earnings announcements, respectively. Comparing these two return series, we notice that the increase at the event interval 0 appears more substantial in response to overnight earnings announcements than to daytime earnings announcements. Similarly, the pattern of returns following overnight earnings announcements is smoother than the pattern of returns following daytime earnings announcements. Together with the results in Table 8, this finding suggests that although the price adjustments to both daytime and overnight earnings announcements occur very rapidly, the overnight period appears to provide investors with an important real time lag to evaluate the information content of overnight earnings news. The finding also suggests that in terms of trading time, information from overnight earnings announcements is incorporated into prices more quickly upon the commencement of

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trading. By comparison, investors take several intervals to evaluate and absorb the information content of a daytime earnings announcement, which is also followed by greater fluctuations in the cumulative portfolio returns in fifteen 15-minute intervals after the event. This finding is similar to the evidence based on US data (Greene and Watts, 1996; Masulis and Shivakumar, 2002), despite the difference in market structure between the ASX and the NYSE and NASDAQ.

Insert Figures 5 and 6 here

Overall, by comparing the price discovery process with respect to earnings announcements made during the pre-opening period versus the normal trading period, we show that the former represent more than noise alone. To some extent, information released before and during the pre-opening period is incorporated more quickly into stock prices upon the commencement of trading relative to information arriving during normal trading hours.

5. Conclusions

A trading day at the ASX consists primarily of the pre-opening, normal trading and closing periods, all of which can play an important role in the price discovery process. Our study focuses on the pre-opening period. Although the pre-opening period involves no actual trading, it can still facilitate the process in which overnight information is incorporated into the opening prices through via order submission. Given the growing trend of firms to release their announcements during the close-to-open period, we study the impact of overnight information arrival on pre-opening price discovery and order placement activities.

Our results show that the efficiency of indicative opening prices gradually increases from 9:30 am up to the opening time. More importantly, this pattern becomes more significant when overnight announcements are released. Additionally, the intensity of overnight announcements influences order placement activities during the pre-opening

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period. When comparing returns following overnight and daytime earnings announcements, we find that price adjustments in response to overnight announcements occur primarily within the overnight period, with limited spillover effects after the market opens. By comparison, daytime earnings announcements can induce price fluctuations over a longer post-event trading period. Thus, the practice of providing investors with ample time to digest new information before trading commences appears to improve rather than hamper the price discovery process.

These findings imply that despite the non-binding nature of pre-opening orders, such orders do not simply reflect noise trading and/or manipulation activities, for returns over a trading day can be largely predicted by close-to-open returns. Most importantly, given that this relationship can be linked to overnight information arrivals, these findings provide strong support for the hypothesis that substantial investor learning can occur even in the absence of trading. Taken together, our research demonstrates that the pre-opening period is a very important channel for facilitating the price discovery process.

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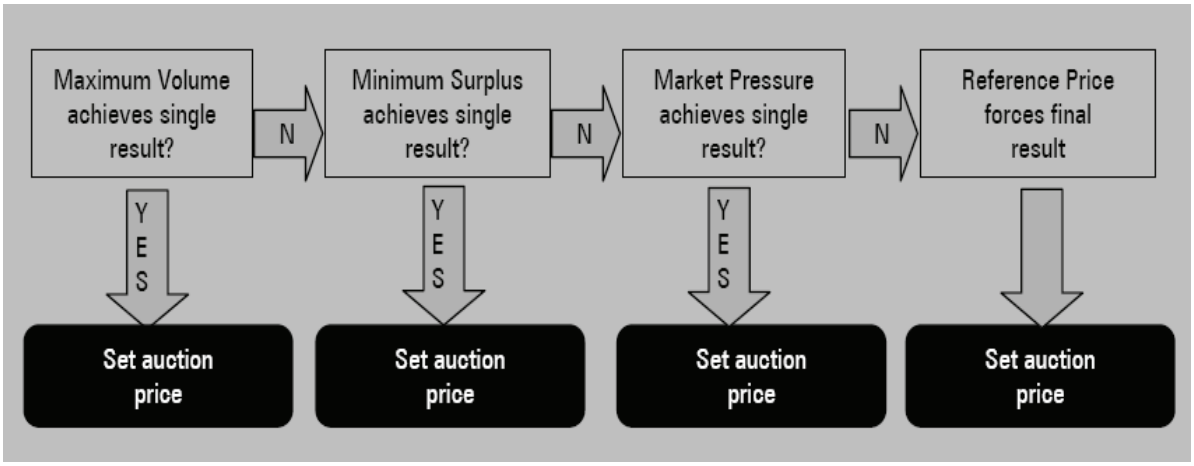
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Appendix A: Auction algorithm based on four principles in the ASX



This figure presents 4 principles for the ASX's opening price algorithm. Principle 1 maximizes the executed trading volume. If a single price cannot be reached, then Principle 2 selects the price at which the surplus is minimized. If a single price cannot be reached with minimum surplus, then Principle 3 selects the highest (lowest) price if the market pressure is on the buy (sell) side. Finally, if a single price cannot be reached after taking market pressure into account, then Principle 4 sets a single opening price based on the reference price.

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Table 1: Summary statistics of stock returns

	Sample		No-news Sample		News Sample	
	R_Ind	R_Close	R_Ind	R_Close	R_Ind	R_Close
MEAN	0.015	0.050	0.018	0.046	0.006	0.062
STD	0.013	0.015	0.012	0.014	0.014	0.017
MIN	-0.503	-0.145	-0.503	-0.108	-0.496	-0.145
MAX	0.481	0.077	0.481	0.077	0.419	0.074

This table presents the descriptive statistics regarding the indicative returns (R_Ind) and the close-to-close returns (R_Close) for the entire sample, the subsample with at least one overnight news announcement (News sample), and the subsample without any overnight news announcements (No-news sample). The indicative return is the natural logarithm of the minute indicative opening prices over the previous closing price. The close-to-close return is the natural logarithm of the closing price over the previous closing price. Mean returns are scaled by 100.

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Table 2: Descriptive statistics for order placement during the pre-opening period

Aggressiveness	Time	Submit	Forward	Same	Backward	Delete
3	7:00-9:30	17.41	3.52	7.07	2.53	10.69
3	9:30-Open	35.05	51.7	52.54	51.82	26.46
3		52.46	55.22	59.6	54.35	37.15
2	7:00-9:30	12.85	4.24	6.59	3.83	17.26
2	9:30-Open	13.49	31.07	15.03	32.49	17.02
2		26.34	35.3	21.63	36.32	34.28
1	7:00-9:30	18.74	7.38	16.41	6.48	25.37
1	9:30-Open	2.47	2.10	2.35	2.85	3.19
1		21.20	9.48	18.76	9.33	28.56

This table reports the average proportion of orders among all orders at a particular aggressiveness level during the market pre-opening period. Level 3 orders are buy (sell) orders with prices higher (lower) than or equal to the indicative opening price. Level 2 orders are buy (sell) orders with prices that are higher (lower) than or equal to the best ask (bid) prices and lower (higher) than the indicative opening price. Level 1 orders are buy (sell) orders with prices lower (higher) than the best ask (bid) prices.

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Table 3: The proportion and the size of orders during the pre-opening period

Panel A: By Time	Number of orders (%)	Order Size	Time	Number of orders (%)	Order Size
7:00-7:30	11.74	5532	9:00-9:10	3.28	36491
7:30-8:00	2.56	18576	9:10-9:20	3.33	42717
8:00-8:30	8.35	5943	9:20-9:30	4.24	14461
8:30-9:00	6.01	44078	9:30-9:40	5.49	12979
9:00-9:30	7.73	29926	9:40-9:50	10.58	11999
9:30-Open	63.61	16100	9:50-Open	73.09	16579
	100.00			100.00	
Panel B: By Order Type	7:00 am – Open			9:30 am – Open	
Order Type	Number of orders (%)	Order Size		Number of orders (%)	Order Size
Submit	49.81	14908		25.26	9802
Revise	50.19	20305		38.35	25721
	100			63.61	
Forward	20.66	50827		17.98	62170
Same	2.16	12415		1.51	13996
Backward	16.23	6698		13.70	8632
Delete	11.14	15281		5.16	18089
	50.59			38.35	

This table reports the overall frequency distribution of each type of orders during the pre-opening period. *Submit* orders are either new buy or new sell orders entering the trading system. *Revise* orders are orders that are revised during the pre-opening period, which includes *Forward, Same, Backward, Delete* orders. *Forward* orders are orders adjusted to prices that are more aggressive than their previously submitted prices (i.e., higher prices for buy orders or lower prices for sell orders). *Same* orders are orders that are amended (increased or decreased) in terms of order size with no change in order price. *Backward* orders are orders that are adjusted to prices that are less aggressive (i.e., lower prices for buy orders or higher prices for sell orders). *Delete* orders are orders that are deleted from the trading system. *Order Size* is the mean order size measured by the number of shares. Panel A is categorized by time and Panel B is categorized by the type of order.

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Table 4: Characteristics of the overlapping market during the pre-opening period

By Time	Frequency (%)			Average Size (AUD)		
Time	Positive Spread	Lock Spread	Cross Spread	Positive Spread	Lock Spread	Cross Spread
7:00-7:30	2.66	1.29	10.97	0.08	0	0.39
7:30-8:00	0.12	0.09	2.22	0.21	0	0.88
8:00-8:30	0.19	0.2	7.14	0.13	0	1.02
8:30-9:00	0.04	0.03	5.51	0.06	0	1.88
9:00-9:30	0.02	0.02	7.08	0.07	0	2.27
9:30-Open	0.02	0.01	62.37	0.03	0	2.76
Total	3.06	1.65	95.3			

This table reports the characteristics of the spread during the market pre-opening period at half-hour intervals. “Positive spread” occurs when the best bid is smaller than the best ask. “Lock spread” occurs when the best bid is equal to the best ask. “Cross spread” occurs when the best bid is greater than the best ask. Frequency is the proportion (%) of each type of spreads among all spread types at each interval.

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Table 5: The weighted price contribution (WPC)

Weighted Price Contribution	Pre-Close to 9:30 am	9:30 am to Open	Open to Close
Mean	0.17	0.14	0.69
Standard Error	0.10	0.06	0.10

This table presents the weighted price contribution (*WPC*) to the close-to-close price change. The close-to-close price change is measured for three periods: the close of the previous trading day to 9:30 am, 9:30 am to the opening of the trading day, and the opening to the closing of the trading day. The *WPC* is calculated stock by stock and then averaged across all stocks for each of the periods.

$$WPC_i = \sum_{t=1}^{T} \left(\frac{\left|\Delta P_{close,t}\right|}{\sum_{t=1}^{T} \left|\Delta P_{close,t}\right|} \right) \left(\frac{\Delta P_{i,t}}{\Delta P_t} \right)$$

where WPC_i is the weighted price contribution of period i to the daily price change. The whole first term in parenthesis on the right hand side of the WPC_i formula is the weighting factor for each day in the sample period, ΔP_t is the daily price change for day t from the previous closing price to the trading day closing price, and $\Delta P_{i,t}$ is the total price change for period i on day t.

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Table 6: Determinants of order placement during the pre-opening period

Order Type		*N_News*	*R_NYSE*	*Depth Same*	*Depth Oppo*	*Vola*	*Size*	*Cross*
Panel A:	Estimates	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Submit	Coef	0.01	0.25	0.07	-0.05	0.35	0.03	-0.18
	t-stat	2.58***	2.94***	9.70***	-7.82***	1.69*	4.58***	-8.84***
Forward	Coef	0.05	-0.13	0.18	-0.02	0.60	0.08	-0.34
	t-stat	2.24**	-2.61***	9.33***	-5.35***	1.58	7.43***	-5.55***
Same	Coef	0.03	-0.26	0.23	-0.02	1.01	-0.13	-0.44
	t-stat	1.76*	-1.65*	4.13***	-1.88*	1.73*	-4.94***	-4.54***
Backward	Coef	-0.07	-0.48	0.15	-0.04	0.26	-0.01	-0.31
	t-stat	-2.15**	-1.98**	10.72***	-2.76***	1.82*	-2.45**	-6.24***
Delete	Coef	0.03	-0.07	0.17	-0.05	1.30	-0.01	-0.33
	t-stat	1.81*	-2.81***	6.26***	-2.86***	1.78*	-1.78*	-5.56***
Panel B: Marginal Effects								
	Aggressive Level							
Submit	3	0.38	3.61	2.01	-1.27	10.55	0.68	-5.51
	2	-0.16	-0.82	-0.41	0.27	-2.43	-0.14	1.07
	1	-0.12	-2.79	-1.60	0.99	-8.12	-0.54	4.43
Forward	3	0.20	-3.23	4.70	-0.56	7.18	2.54	-3.76
	2	-0.06	2.19	-3.30	0.36	-3.18	-1.91	1.82
	1	-0.14	1.03	-1.40	0.20	-3.99	-0.62	1.94
Same	3	0.35	3.10	3.00	0.04	10.04	-2.87	-7.50
	2	-0.08	-2.33	-1.02	0.02	-3.08	1.02	2.69
	1	-0.26	-0.77	-1.99	-0.07	-6.95	1.85	4.81
Backward	3	-0.80	-6.94	4.65	-0.10	3.02	-0.63	-3.64
	2	0.38	3.42	-2.72	0.14	-0.01	0.21	0.80
	1	0.42	3.52	-1.93	-0.04	-3.01	0.42	2.84
Delete	3	0.22	-2.37	3.06	-0.44	12.66	-0.49	-6.22
	2	-0.01	-1.26	-0.08	0.02	-0.65	-0.16	-0.51
	1	-0.21	3.63	-2.98	0.43	-12.01	0.65	6.73

This table reports the determinants of order placement activities during the market pre-opening period. We estimate the regression equations (4a) and (4b) for each stock and report the cross-sectional averages of coefficient estimates. The dependent variable is the level of order aggressiveness for each type of orders. N_News is the intensity of overnight news announcements, which is measured by the cumulative number of company announcements released between the previous closing day and the time of order submission or revision. R_NYSE is the NYSE Composite Index return of the previous trading day. $Depth_Same$ ($Depth_Oppo$) is the natural logarithm of the total number of accumulated shares for the same (opposite) sides of the order book. $Vola$ is the lag of the absolute value of the close-to-close price change. $Size$ is the natural logarithm of the order size measured by the number of shares. $Cross$ is the absolute value of the difference between the best bid and the best ask at the time of order submission or revision. *Coef* refers to the mean coefficient estimate and *t-stat* refers to the average *t*-statistic. Marginal effects are calculated using the sample average of individual marginal effects and scaled by 100. Panel A presents the coefficient estimates. Panel B presents the marginal effects. The symbols ***, **, and * denote the 1%, 5%, and 10% significance levels, respectively.

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Table 7: The positive and negative earnings announcements

Period	Total		Positive News		Negative News	
	No	Percentage	No	Percentage	No	Percentage
Daytime	114	25%	60	23.6%	54	30%
Overnight	319	75%	194	76.4%	125	70%
Total	423	100%	254	100%	179	100%

This table reports the number and proportion of earnings announcements. Positive (Negative) News is the earnings announcement with the positive (negative) AFE.

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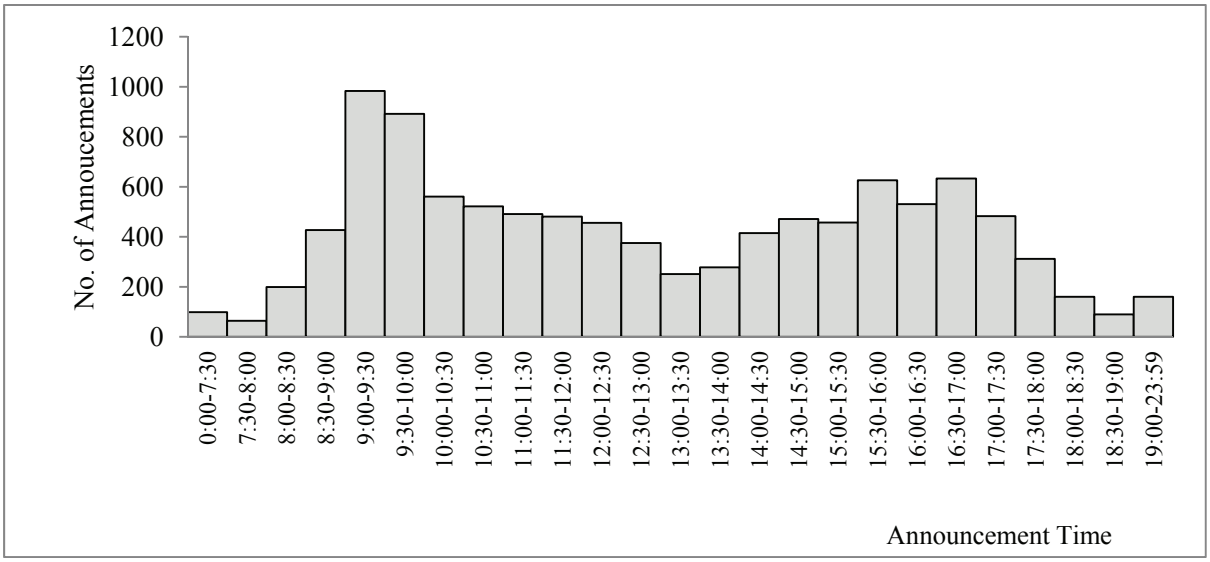
Table 8: Portfolio returns for daytime and overnight earnings announcements

	Daytime Earnings Announcements			Overnight Earnings Announcements		
Event Interval	Mean Return (%)	*t*-stats	Cumulative Return (%)	Mean Return (%)	*t*-stats	Cumulative
-15	0.03	0.98	0.03	0.01	0.38	0.01
-14	-0.01	-0.21	0.02	-0.01	-0.53	0.00
-13	0.05	1.55	0.07	0.02	1.18	0.02
-12	-0.03	-0.66	0.04	0.00	-0.13	0.02
-11	-0.05	-1.12	-0.01	0.00	0.18	0.02
-10	-0.02	-0.48	-0.03	-0.01	-0.93	0.01
-9	-0.05	-1.47	-0.08	0.01	0.83	0.02
-8	0.01	0.20	-0.07	0.01	0.74	0.03
-7	-0.03	-0.82	-0.10	0.00	0.13	0.03
-6	0.04	1.63	-0.06	0.02	1.33	0.05
-5	0.04	0.77	-0.02	-0.01	-0.65	0.04
-4	-0.04	-0.97	-0.06	-0.01	-0.56	0.03
-3	0.02	0.57	-0.04	0.02	0.89	0.05
-2	-0.04	-0.92	-0.08	0.00	0.10	0.05
-1	0.06	1.12	-0.02	0.05	1.22	0.10
0	0.10	2.58***	0.08	0.15	2.74***	0.25
1	0.07	1.73*	0.15	0.04	1.65*	0.29
2	-0.07	-1.79*	0.08	-0.03	-0.95	0.26
3	0.09	2.15**	0.17	0.03	1.05	0.29
4	-0.01	-0.10	0.16	-0.03	-1.17	0.26
5	-0.02	-0.43	0.14	0.02	0.95	0.28
6	0.04	1.17	0.18	-0.01	-0.54	0.27
7	-0.05	-1.45	0.13	0.00	0.05	0.27
8	0.06	1.76*	0.19	-0.03	-1.54	0.24
9	-0.01	-0.18	0.18	-0.01	-0.60	0.23
10	0.01	0.33	0.19	-0.01	-0.42	0.22
11	-0.03	-1.19	0.16	0.00	0.31	0.22
12	0.02	0.89	0.18	0.00	0.29	0.22
13	-0.02	-0.35	0.16	0.01	0.60	0.23
14	0.01	0.32	0.17	0.00	0.13	0.23
15	0.02	0.20	0.19	-0.01	-1.46	0.22

This table reports the portfolio returns around the release time of daytime and overnight earnings announcements on a trading strategy of buying stocks with positive earnings news and selling stocks with negative earnings news. For daytime earnings announcements, we define the event interval 0 as the 15-minute interval during which the earnings announcement is made. For overnight earnings announcements, the event interval 0 starts from the closing time of the previous trading day and ends at the opening time of the trading day (or at the first trade if trading halts occur). The other event intervals are defined relative to the event interval 0: fifteen 15-minute intervals prior and fifteen 15-minute intervals subsequent to the event interval 0. The *t*-statistics represent the test of the null hypothesis of the zero mean portfolio return. The symbols ***, **, and * denote the 1%, 5%, and 10% significance levels, respectively.

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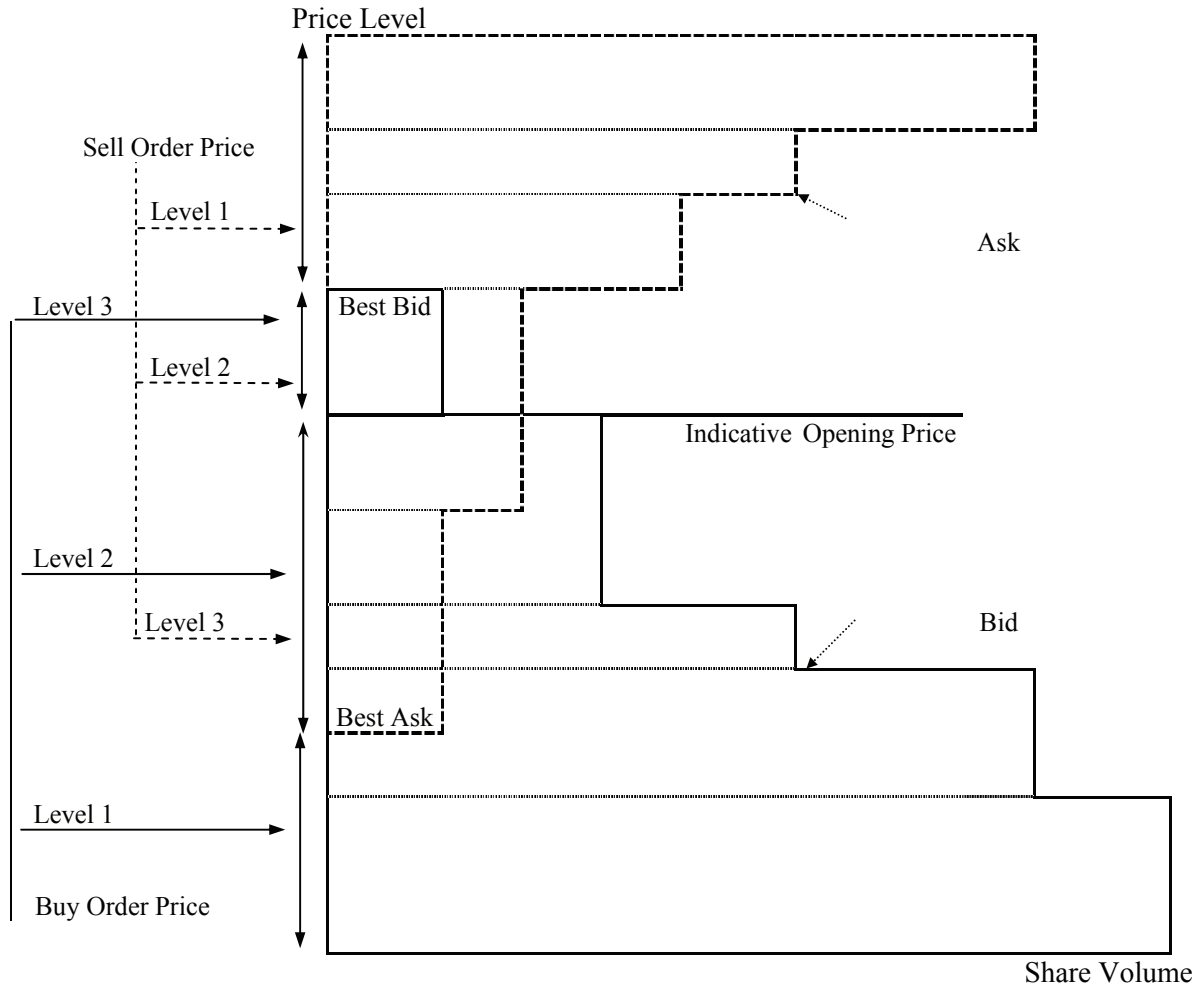
Figure 1: Frequency distribution of news announcements at half-hour intervals



This figure presents the distribution of the total number of news announcements in our sample at half-hour intervals. The sample consists of 10,417 news announcements, with 5,222 announcements (50.13%) released during the overnight period and 5,195 announcements (49.87%) released during normal trading hours.

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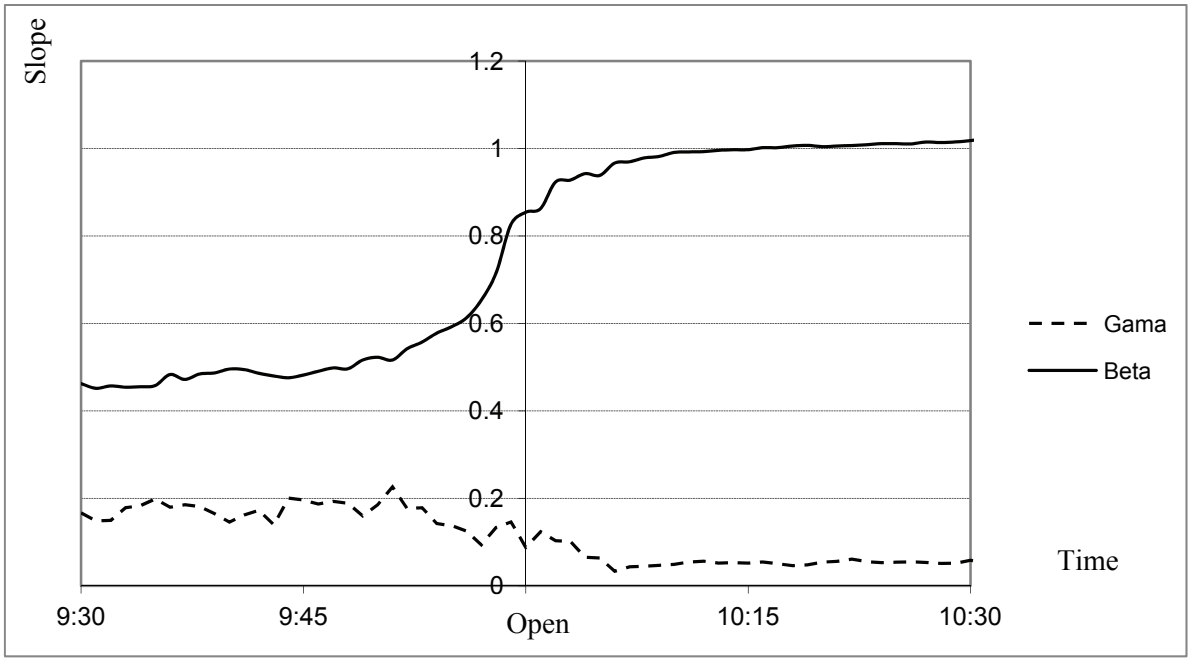
Figure 2: The levels of order aggressiveness during the pre-opening period



This figure illustrates the limit order book and the classification of order aggressiveness during the pre-opening period. The book overlaps in the sense that the best bid price exceeds the best ask price during this period. The dashed line refers to the order book on the sell (ask) side and the solid line denotes the order book on the buy (bid) side. The indicative opening price is the opening price following the opening algorithm of the ASX if the market were to open at the order submission time. Orders are categorized into three levels of aggressiveness corresponding to the probability that they would be executed if the market were to open at that time. The least aggressive orders (Level 1) are buy (sell) orders with prices that are lower (higher) than the best ask (bid) prices. Level 2 orders are buy (sell) orders with prices that are higher (lower) than or equal to the best ask (bid) prices and lower (higher) than the indicative opening price. The most aggressive orders are Level 3 orders, which are buy (sell) orders with prices higher (lower) than or equal to the indicative opening price.

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Figure 3: Overnight news announcements and price efficiency



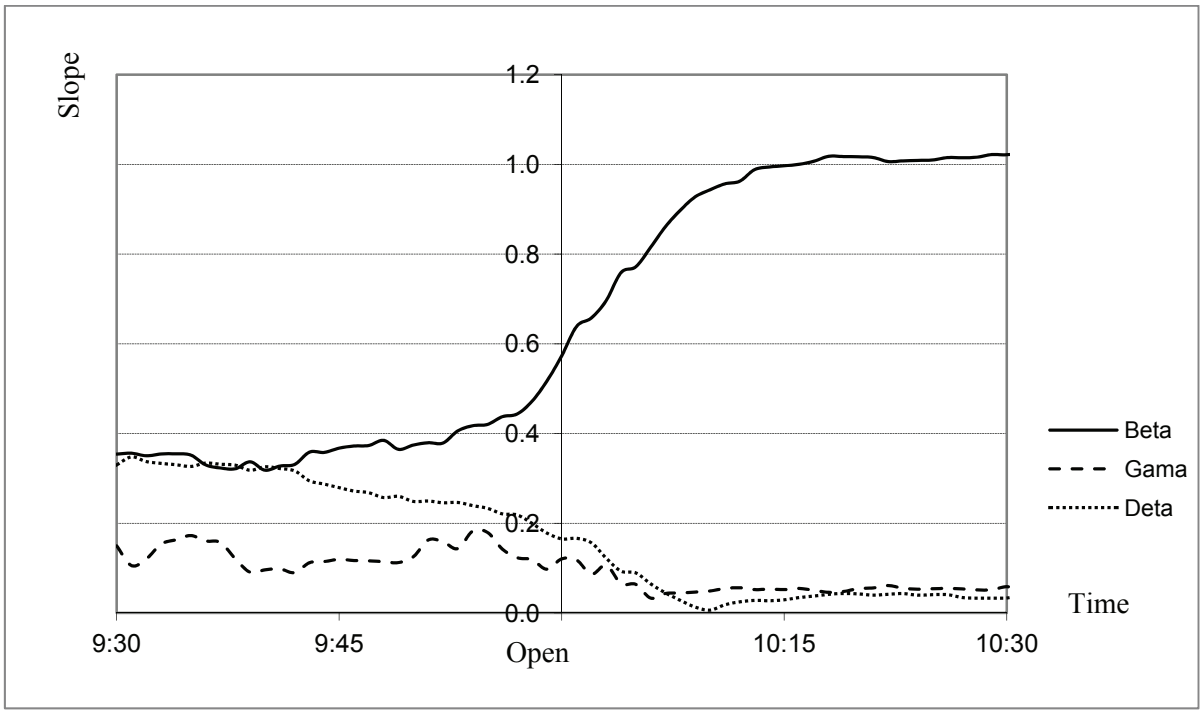
This figure presents the mean coefficient estimates for each one-minute interval from 9:30 am to 10:30 am. We estimate the following regression (2) for each one-minute interval across the sample period and stock by stock and report the mean coefficients.

$$R_Close = \alpha + \beta * R_Ind_t + \gamma * D_News_t * R_Ind_t + \varepsilon_t \quad (2)$$

where R_Close is the close-to-close return, which is equal to the natural logarithm of the closing price over the previous closing price. R_Ind_t is the indicative return, which is equal to the natural logarithm of the indicative opening price (or the traded price after the market opens) at time *t* over the previous close price. The news dummy (D_News_t) equals one if at least one announcement arrives between the closing time of the previous trading day and time *t* and equals zero if otherwise. The solid line is the mean coefficient estimate of β and the dashed line is the mean coefficient estimate of γ. All coefficient estimates are reported at the 5% level of significance.

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Figure 4: Overnight news announcements, the NYSE Composite Index, and price efficiency



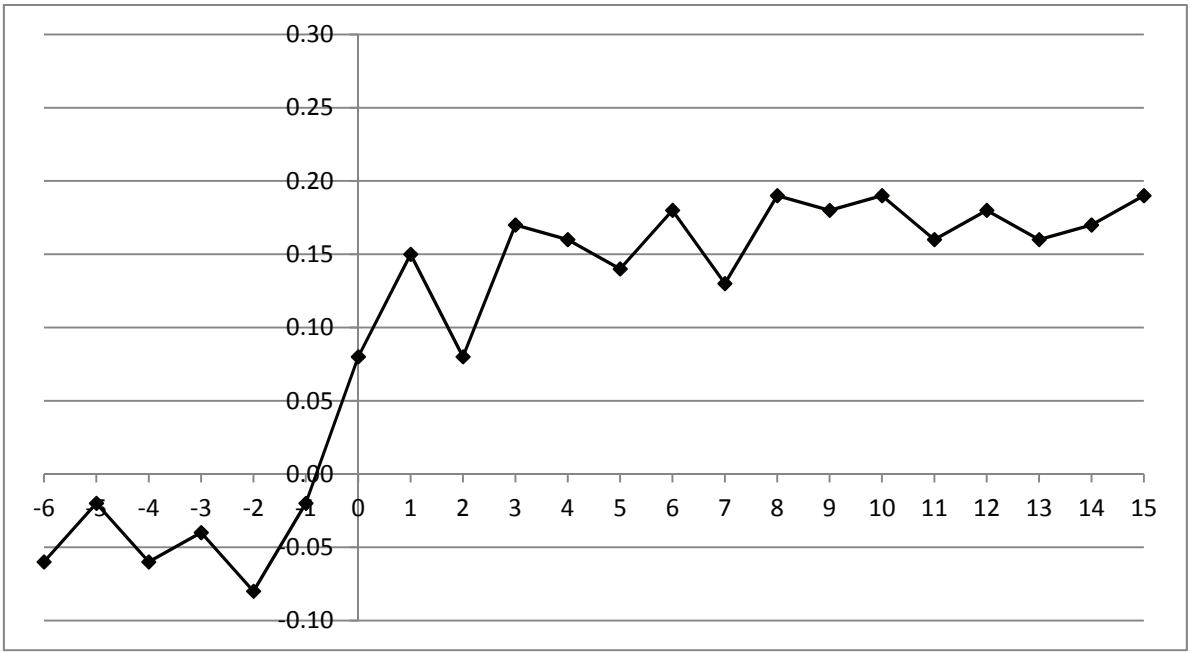
The figure presents the mean coefficient estimates for each one-minute interval from 9:30 am to 10:30 am. We estimate the following regression (3) for each one-minute interval across the sample period and stock by stock and report the mean coefficients.

$$R_Close = \alpha + \ \delta * R_NYSE + \beta * R_Ind_t + \gamma * D_News_t * R_Ind_t + \varepsilon_t \quad (3)$$

where R_Close is the close-to-close return, which is equal to the natural logarithm of the closing price over the previous closing price. R_NYSE is the return of the NYSE Composite Index from the previous trading day, which represents the flow of overnight and overseas information arriving during the pre-opening period. R_Ind_t is the indicative return, which is equal to the natural logarithm of the indicative opening price (or the traded price after the market opens) at time *t* over the previous close price. The news dummy (D_News_t) equals one if at least one announcement arrives between the closing time of the previous trading day and time *t* and equals zero if otherwise. The dotted line is the mean coefficient estimate of δ, the solid line is the mean coefficient estimate of β, and the dashed line is the mean coefficient estimate of γ. All coefficient estimates are reported at the 5% level of significance.

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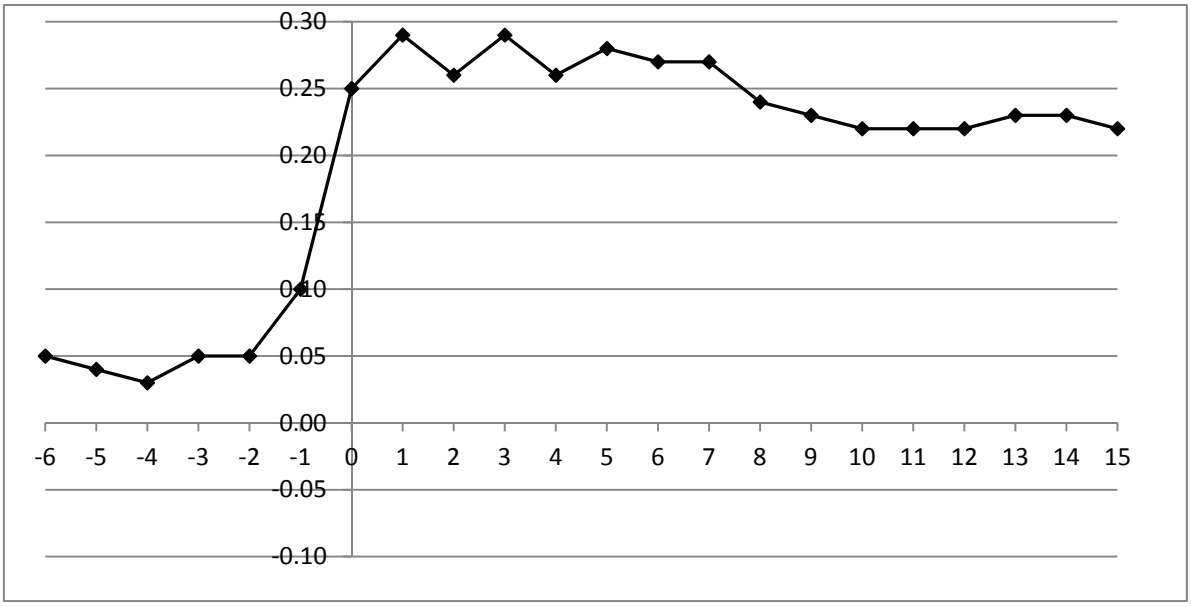
Figure 5: The cumulative portfolio returns for daytime earnings announcements



This figure shows the cumulative portfolio returns of six 15-minute intervals prior and fifteen 15-minute intervals subsequent to the event interval 0 for a trading strategy of buying stocks with positive earnings news and selling stocks with negative earnings news. All earnings announcements are released during normal trading hours. The event interval 0 is the 15-minute interval during the normal trading hours at which the earning announcements are made.

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Figure 6: The cumulative portfolio returns for overnight earnings announcements



This figure reports the cumulative portfolio returns of six 15-minute intervals prior to and fifteen 15-minute intervals subsequent to the event interval 0 (the announcement interval) for a trading strategy of buying stocks with positive earnings news and selling stocks with negative earnings news. All earnings announcements are released during the overnight period. The event interval 0 starts from the closing time of the previous trading day and ends at the opening time of the trading day (or at the first trade if trading halts occur).

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